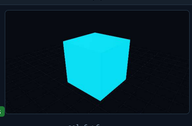
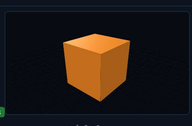
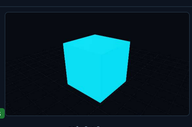
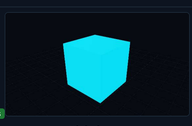
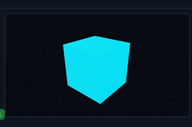
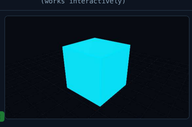
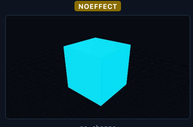
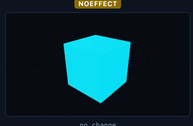
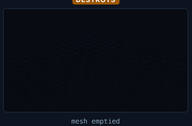
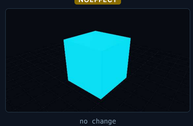
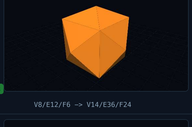
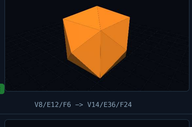
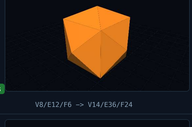
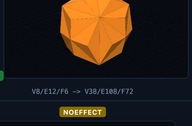
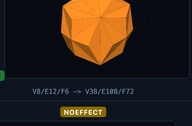
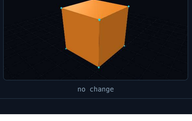
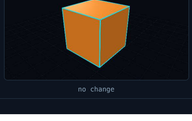
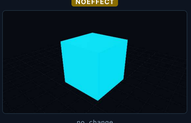
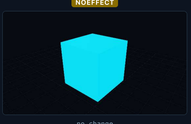
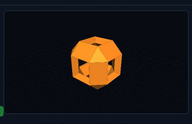
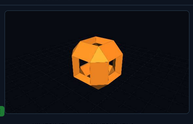
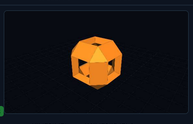
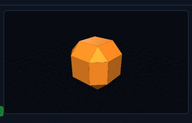
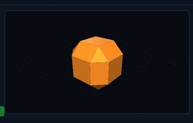
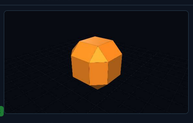
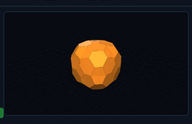
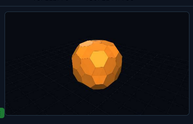
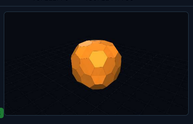
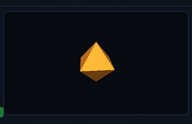
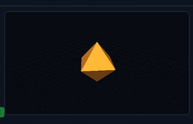
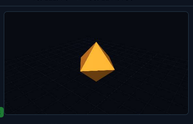
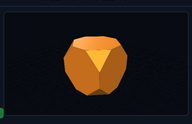
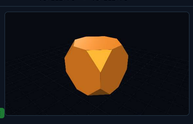
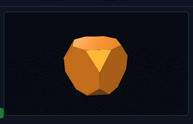
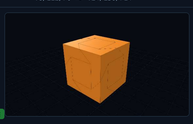
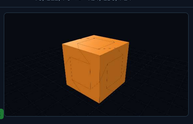
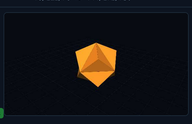
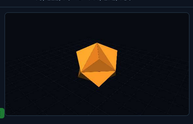
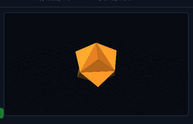
TOPMOD 3D — Tool × Modifier compatibility matrix

Each cell = create a **Cube** → switch to the modifier (selection mode) → **Select All** → apply the tool. 93 combinations.

DESTROYS 2 · **NOEFFECT** 22 · **PASS** 54 · **UNVERIFIED** 15

Tool \ Modifier	Verts (1)	Edges (2)	Faces (3)
Select All ⌘A `selAll`	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	 PASS sel 6->6
Select Inverse `selInv`	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	 PASS sel 6->0
Select Similar `selSimilar`	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	 PASS sel 6->6
Grow Selection `selGrow`	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	 PASS sel 6->6
Shrink Selection `selShrink`	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	 PASS sel 6->6
Select Edge Loop `edgeLoop`	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	 PASS sel 6->6
Select Edge Ring `edgeRing`	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	 PASS sel 6->6
Select Face Loop `faceLoop`	UNVERIFIED  selection-only tool; harness tracks face-selection only (works interactively)	 PASS sel 0->6	 PASS sel 6->0
Insert Edge `insertEdge`	NOEFFECT  no change	NOEFFECT  no change	NOEFFECT  no change
Delete Edge `deleteEdge`	NOEFFECT  no change	NOEFFECT  no change	NOEFFECT  no change
Delete Face `deleteFace`	NOEFFECT  no change	NOEFFECT  no change	DESTROYS mesh emptied
Delete Vertex `deleteVertex`	DESTROYS  mesh emptied	NOEFFECT  no change	NOEFFECT  no change
Subdivide Edge `subdivEdge`	NOEFFECT  no change	 PASS V8/E12/F6 -> V20/E24/F6	NOEFFECT  no change
Triangulate `triangulate`	 PASS V8/E12/F6 -> V14/E36/F24	 PASS V8/E12/F6 -> V14/E36/F24	 PASS V8/E12/F6 -> V14/E36/F24
Stellate `stellate`	 PASS V8/E12/F6 -> V14/E36/F24	 PASS V8/E12/F6 -> V14/E36/F24	 PASS V8/E12/F6 -> V14/E36/F24
Double Stellate `doubleStellate`	 PASS V8/E12/F6 -> V38/E108/F72	 PASS V8/E12/F6 -> V38/E108/F72	 PASS V8/E12/F6 -> V38/E108/F72
Extrude Face `extrude`	NOEFFECT  no change	NOEFFECT  no change	 PASS V8/E12/F6 -> V32/E60/F30

Tool \ Modifier	Verts (1)	Edges (2)	Faces (3)
Dome Extrude `dome`	NOEFFECT  no change	NOEFFECT  no change	 PASS V8/E12/F6 -> V86/E180/F96
Fill Holes `fillholes`	NOEFFECT  no change	NOEFFECT  no change	NOEFFECT  no change
Add Handle (2 faces) `handle`	NOEFFECT  no change	NOEFFECT  no change	NOEFFECT  no change
Create Rind / Crust `crust`	 PASS V8/E12/F6 -> V16/E24/F12	 PASS V8/E12/F6 -> V16/E24/F12	 PASS V8/E12/F6 -> V16/E24/F12
Wireframe Modeling `wireframe`	 PASS V8/E12/F6 -> V48/E120/F64	 PASS V8/E12/F6 -> V48/E120/F64	 PASS V8/E12/F6 -> V48/E120/F64
Menger Sponge `menger`	 PASS V8/E12/F6 -> V896/E2112/F1056	 PASS V8/E12/F6 -> V896/E2112/F1056	 PASS V8/E12/F6 -> V896/E2112/F1056
Catmull-Clark `catmull`	 PASS V8/E12/F6 -> V26/E48/F24	 PASS V8/E12/F6 -> V26/E48/F24	 PASS V8/E12/F6 -> V26/E48/F24
Doo-Sabin `doosabin`	 PASS V8/E12/F6 -> V24/E48/F26	 PASS V8/E12/F6 -> V24/E48/F26	 PASS V8/E12/F6 -> V24/E48/F26
Loop `loop`	 PASS V8/E12/F6 -> V50/E144/F96	 PASS V8/E12/F6 -> V50/E144/F96	 PASS V8/E12/F6 -> V50/E144/F96
Honeycomb `honeycomb`	 PASS V8/E12/F6 -> V96/E144/F50	 PASS V8/E12/F6 -> V96/E144/F50	 PASS V8/E12/F6 -> V96/E144/F50
Dual `dual`	 PASS V8/E12/F6 -> V6/E12/F8	 PASS V8/E12/F6 -> V6/E12/F8	 PASS V8/E12/F6 -> V6/E12/F8
Vertex Truncation `truncate`	 PASS V8/E12/F6 -> V24/E36/F14	 PASS V8/E12/F6 -> V24/E36/F14	 PASS V8/E12/F6 -> V24/E36/F14
Pentagonalization `pentagonalize`	 PASS V8/E12/F6 -> V44/E72/F30	 PASS V8/E12/F6 -> V44/E72/F30	 PASS V8/E12/F6 -> V44/E72/F30
√3 Subdivision `root3`	 PASS V8/E12/F6 -> V14/E36/F24	 PASS V8/E12/F6 -> V14/E36/F24	 PASS V8/E12/F6 -> V14/E36/F24